OpenELIS Instrument Integration — Software & Export Companion Guide

Purpose: Software download links, step-by-step export instructions, LIS configuration guides, and published validation datasets for each instrument integrated with OpenELIS Global.
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Maintained by: OpenELIS Global Community

Table of Contents

1. QuantStudio 5 / 7 Flex (Real-Time PCR)
2. MALDI Biotyper / Sirius (Organism ID)
3. Attune CytPix (Flow Cytometry)
4. MinION + TB-Profiler (WGS TB Resistance)
5. FluoroCycler XT (RT-PCR / Line Probe Assays)
6. SeqStudio Genetic Analyzer (Sanger Sequencing)
7. KingFisher Flex (Extraction Robot)

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1. QuantStudio 5 / 7 Flex

Instrument Overview

Item	Detail
Manufacturer	Thermo Fisher Scientific / Applied Biosystems
Models	QuantStudio 5 (96-well), QuantStudio 7 Flex (96/384-well)
Use Cases	HIV Viral Load, TB Detection, Respiratory Panels, COVID-19
Export Format	Tab-delimited CSV from Design & Analysis Software

Software Downloads

Software	Version	Link	Notes
QuantStudio Design &	2.7.x	ThermoFisher Connect	Requires ThermoFisher Connect account

Software	Version	Link	Notes
Analysis Software (desktop)		Downloads	(free). Desktop application for Windows.
QuantStudio Design & Analysis Software (cloud)	2.x	ThermoFisher Connect Cloud	Browser-based version. Same export functionality. Login required.
Instrument Firmware	Varies	ThermoFisher Connect → Instruments	Firmware updates for the instrument itself.

Registration note: All ThermoFisher software downloads require a free ThermoFisher Connect account. Register at thermofisher.com/connect.

Export Instructions — Results CSV

Menu Path: Analysis → Export → Results

1. Open your completed experiment in QuantStudio Design & Analysis Software
2. Ensure analysis is complete (green checkmark on all wells or "Analysis" tab shows results)
3. Navigate to **Export** tab (top menu bar)
4. In the Export dialog:
   - **Content:** Select `Results`
   - **File Type:** Select `CSV` (or `.txt` — both produce tab-delimited output)
   - **Fields to include:** Ensure at minimum: `Well`, `Well Position`, `Sample Name`, `Target Name`, `Task`, `Reporter`, `Quencher`, `CT`, `Ct Mean`, `Ct SD`, `Quantity`, `Quantity Mean`, `Quantity SD`, `Amp Status`, `Amp Score`
   - For Standard Curve experiments also include: `Standard Quantity`, `Automatic Ct Threshold`, `Ct Threshold`
5. Click **Export**
6. Choose save location → the file is generated with metadata header block + `[Results]` section

Video Guides:

- ThermoFisher: Exporting Data from QuantStudio Software — Check the "Training & Support" tab for video tutorials
- Applied Biosystems YouTube: QuantStudio Software Walkthrough — Search for "QuantStudio Design and Analysis export"

Key export settings to verify:

- Ensure `Sample Name` contains the OpenELIS accession number

- Confirm `Amp Status` column is included (some export presets omit it)
- For quantitative assays (HIV VL), ensure `Quantity` column is present and the standard curve has been applied

Published Validation Data & References

Resource	Link	Description
ThermoFisher QuantStudio Application Notes	thermofisher.com/quantstudio-resources	Application notes with example datasets for various assay types
WHO HIV VL EQA Program	QCMD	Quality Control for Molecular Diagnostics — proficiency panels with known copy numbers
CDC DGHT Molecular QA Resources	CDC International Laboratory Branch	Reference materials for HIV VL and EID testing
Abbott RealTime HIV-1 Package Insert	Provided with reagent kit	Contains expected Ct ranges and standard curve parameters for validation

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2. MALDI Biotyper / Sirius

Instrument Overview

Item	Detail
Manufacturer	Bruker Daltonics
Models	MALDI Biotyper, MALDI Biotyper Sirius
Use Cases	Microbial organism identification (bacteria, fungi, mycobacteria)
Export Formats	ASTM E1394 over TCP/IP (primary), CSV export (fallback)

Software Downloads

Software	Version	Link	Notes
MBT Compass (IVD)	4.x / 6.x	Bruker Clinical Microbiology Portal	Contact Bruker representative or download via Bruker Care portal. IVD-certified version.

Software	Version	Link	Notes
MBT Compass (RUO)	4.x / 6.x	Same portal	Research Use Only — larger database, more configurable.
MBT Compass IVD Library	Rev 12+	Provided by Bruker with service contract	Library updates require Bruker service contract. Includes IVD bacteria, mycobacteria, and fungi modules.
flexControl	4.x	Bruker Downloads	Instrument control software (for hardware operation, not results).

Registration note: MBT Compass downloads require a Bruker Care account and active service/support contract. Contact your Bruker field service representative for access.

Export Instructions — CSV Results Table

Menu Path: MBT Compass → Classification → Export Results

1. In MBT Compass, complete the identification run
2. Navigate to the **Classification** view (shows all identified spots)
3. Select all results (Ctrl+A) or use the checkbox column
4. Click **Export** (toolbar button or File → Export → Results Table)
5. In the Export dialog:
   - **Format:** CSV (tab-delimited)
   - **Fields:** Ensure all columns are selected, especially: Analyte Name, Analyte ID, Target Position, Organism (Best Match), Score Value (Best Match), Organism (Second Best Match), Score Value (Second Best Match), Category, Identification, Comment, Analysis Date, Operator, Instrument, Library, Method, Spot Type
6. Save file to the designated import folder

LIS Configuration — ASTM E1394 TCP/IP (Recommended)

Menu Path: MBT Compass → Administration → LIS Configuration

1. Open MBT Compass as Administrator
2. Navigate to **Administration → LIS Configuration**
3. Configure the following:
   - **Connection Type:** TCP/IP
   - **Host/IP Address:** OpenELIS server IP address
   - **Port:** 9100 (or as configured in OpenELIS ASTM listener)

- **Protocol:** ASTM E1394 / LIS2-A2
- **Direction:** Bidirectional (enable both Send Results and Receive Orders)
- **Send Results Automatically:** Enable if desired (auto-sends after each classification)

4. **Field Mapping** (within LIS Configuration):
   - **Sample ID field:** Map to Analyte Name (this carries the accession number)
   - **Result field:** Map to Identification (reported organism name)
   - **Score field:** Enable sending score in Comment records
5. Click **Test Connection** to verify connectivity
6. Run a BTS (Bacterial Test Standard) to verify end-to-end data flow

Key LIS configuration notes:

- The exact field positions in ASTM records depend on this configuration screen. If OpenELIS receives unexpected data placement, check here first.
- The "Test LIS Connection" button sends a sample ASTM message — capture this with a TCP sniffer to validate the message format before deploying the parser.

Video Guides:

- Bruker MALDI Biotyper Training Videos — Check "Resources" tab for video library
- Bruker provides on-site training as part of instrument installation; request LIS integration training specifically

Published Validation Data & References

Resource	Link	Description
Bruker IVD Library Validation Data	Provided with library purchase	Species-level validation data per library module (bacteria, fungi, mycobacteria)
MALDI Biotyper Application Notes	bruker.com/microbiology	Published application notes with identification accuracy studies
WHO EQAS for Microbiology	WHO External Quality Assessment	Proficiency testing programs for clinical microbiology
Singhal et al. (2015) "MALDI-TOF mass spectrometry: an emerging technology for microbial identification and diagnosis"	PubMed: PMC4525074	Comprehensive review of MALDI-TOF validation studies

Resource	Link	Description
Croxatto et al. (2012) "Applications of MALDI-TOF mass spectrometry in clinical diagnostic microbiology"	PubMed: PMC3232461	Large-scale validation across multiple clinical sites
S. pneumoniae / S. mitis Discrimination	PubMed: PMC3553912	Validation data on known MALDI-TOF limitation for viridans streptococci

3. Attune CytPix

Instrument Overview

Item	Detail
Manufacturer	Thermo Fisher Scientific / Invitrogen
Models	Attune NxT, Attune CytPix
Use Cases	CD4/CD8 T-cell enumeration (HIV monitoring), immunophenotyping
Export Format	CSV from Attune Cytometric Software

Software Downloads

Software	Version	Link	Notes
Attune Cytometric Software	5.x	ThermoFisher Connect Downloads	Requires ThermoFisher Connect account. Windows application.
Attune Performance Tracking Software	Included	Bundled with Attune Cytometric Software	QC bead tracking and performance monitoring.

Registration note: Same ThermoFisher Connect account as QuantStudio software.

Export Instructions — Statistics CSV

Menu Path: Experiment → Export → Statistics Table

1. In Attune Cytometric Software, complete your acquisition and analysis

2. Ensure gating is applied and population statistics are calculated
3. Navigate to **Experiment** menu → **Export**
4. In the Export dialog:
   - **Export Type:** Statistics Table
   - **Format:** CSV
   - **Populations:** Select all populations you want exported. For TBNK panel, ensure at minimum: All Events, Singlets, Lymphocytes, CD3+, CD3+CD4+, CD3+CD8+, CD4/CD8 Ratio
   - **Statistics:** Select: Count, % of Parent, Count/uL (absolute count), and any custom statistics
   - **Include metadata:** Enable (adds header block with instrument, operator, panel info)
5. Click **Export**

Alternative export — Plate Layout Export (96-well autosampler):

1. If using the autosampler, you can also export from the **Plate** view
2. File → Export → Plate Statistics
3. This produces one row per well × population × statistic (same format as individual export but for the full plate)

Key export settings to verify:

- Ensure `Count/uL` statistic is selected — this is the volumetric absolute count (the Attune's key feature)
- Verify that the gating template name matches what the OpenELIS parser expects for population names
- Check that `Sample Name` in the experiment setup contains the OpenELIS accession number

Video Guides:

- ThermoFisher: Attune NxT Software Training — Check "Training" section
- Invitrogen YouTube: Attune Software Tutorials — Playlist covering acquisition, analysis, and export
- ThermoFisher Learning Center: Flow Cytometry — Comprehensive flow cytometry training resources

Published Validation Data & References

Resource	Link	Description
Attune NxT Performance Specifications	thermofisher.com/attune	Official specifications including volumetric counting accuracy
BD FACS CD4 Reference Method Comparison	See instrument validation studies in literature	Comparison of Attune volumetric counting vs BD FACSCount/FACSCalibur for CD4

Resource	Link	Description
WHO Prequalification for CD4 Technologies	WHO PQ Diagnostics	List of WHO-prequalified CD4 testing technologies and performance criteria
PEPFAR CD4 Testing Guidelines	PEPFAR Technical Guidance	CD4 testing quality requirements for PEPFAR-supported programs
Pattanapanyasat et al. — CD4 Volumetric Counting Validation	PubMed — Search: "Attune CD4 volumetric counting validation"	Published validation studies comparing Attune to reference methods
Stabilized Blood Controls	Streck CD-Chex / Beckman Coulter Immuno-Trol	QC control materials with lot-specific CD4/CD8 target ranges

4. MinION + TB-Profiler

Instrument Overview

Item	Detail
Manufacturer (hardware)	Oxford Nanopore Technologies
Manufacturer (pipeline)	London School of Hygiene & Tropical Medicine (open source)
Models	MinION Mk1B, MinION Mk1C, GridION (higher throughput)
Use Cases	TB whole-genome sequencing, drug resistance prediction, lineage typing
Export Format	JSON per-sample + CSV batch summary from TB-Profiler

Software Downloads — Sequencing

Software	Version	Link	Notes
MinKNOW (instrument control)	24.x	Oxford Nanopore Community	Requires ONT Community account (free with instrument purchase). Controls MinION sequencing.
Dorado (basecaller)	0.5.x+	GitHub: nanoporetech/dorado	Open source GPU-accelerated basecaller. Recommended over Guppy (deprecated).

Software	Version	Link	Notes
Guppy (legacy basecaller)	6.x	ONT Community Downloads	Legacy; being replaced by Dorado. Still used in some pipelines.
EPI2ME (cloud analysis)	Current	epi2me.nanoporetech.com	ONT's cloud analysis platform. Includes some TB workflows.

Registration note: MinKNOW and Guppy downloads require an Oxford Nanopore Community account, which is provided with instrument purchase. Dorado is open source on GitHub (no account needed).

Software Downloads — Bioinformatics Pipeline

Software	Version	Link	Notes
TB-Profiler	6.3.0+	GitHub: jodyphelan/TBProfiler	**Open source.** Primary pipeline for TB resistance prediction. Install via conda or pip.
TB-Profiler (conda)	Latest	`conda install -c bioconda tb-profiler`	Easiest installation method.
TB-Profiler (pip)	Latest	`pip install tb-profiler`	Alternative Python package installation.
TB-Profiler (Docker)	Latest	`docker pull quay.io/biocontainers/tb-profiler`	Containerized — recommended for reproducible deployments.
WHO Mutation Catalogue Database	2023.5	Bundled with TB-Profiler; update via `tb-profiler update_tbdb`	WHO 2023 catalogue integrated. Auto-updates.
minimap2 (aligner)	2.26+	GitHub: lh3/minimap2	Dependency of TB-Profiler; installed automatically via conda.
samtools	1.19+	GitHub: samtools/samtools	Dependency.
bcftools	1.19+	GitHub: samtools/bcftools	Dependency.

Running TB-Profiler — Quick Reference

Single sample (FASTQ input):

```bash
```

```
tb-profiler profile \
  --read1 sample_001.fastq.gz \
  --platform nanopore \
  --prefix 2502190001 \
  --threads 8 \
  --csv
```

Batch (multiple samples):

```bash
# Run each sample
for sample in barcode01 barcode02 barcode03; do
  tb-profiler profile \
    --read1 ${sample}/*.fastq.gz \
    --platform nanopore \
    --prefix ${sample} \
    --threads 8
done

# Collate into batch summary CSV
tb-profiler collate --prefix batch_20250219
```

Output locations:

- Per-sample JSON: `results/{prefix}.results.json`
- Batch CSV: `{collate_prefix}.txt` (tab-delimited despite .txt extension)

Export Instructions — Getting Results to OpenELIS

Since TB-Profiler is a command-line tool, "export" means collecting the output files:

1. **Per-sample JSON (preferred):** Copy `results/*.results.json` to OpenELIS import folder
2. **Batch CSV:** Run `tb-profiler collate` then copy the output `.txt` file
3. **Automated pipeline:** Set up a cron job or workflow manager (e.g., Nextflow, Snakemake) to: a. Monitor for new FASTQ files from MinKNOW b. Run TB-Profiler automatically c. Copy results JSON to OpenELIS import directory d. Optionally trigger OpenELIS import via API

Nextflow pipeline option:

Pipeline	Link	Description
nf-core/tbprofile	Check nf-core for community pipelines	Nextflow wrapper for TB-Profiler with QC and reporting
ARTIC-TB	GitHub: artic-network	ARTIC network TB sequencing protocols and pipelines

Published Validation Data & References

Resource	Link	Description
TB-Profiler Publication	Phelan et al., Genome Medicine (2019)	Original TB-Profiler paper with validation data
WHO Catalogue of TB Mutations (2023)	WHO TB Mutation Catalogue	The definitive reference for TB resistance mutations and confidence gradings
WHO Technical Manual on TB WGS	WHO TB WGS Manual	Comprehensive guide on implementing TB WGS in reference laboratories
TB-Profiler Benchmark Dataset	GitHub: jodyphelan/TBProfiler/test_data	Test FASTQ files included in the TB-Profiler repository for validation
CRyPTIC Consortium Dataset	CRyPTIC Project	Large-scale WGS + phenotypic DST dataset (>40,000 isolates) for validation
ReSeqTB Knowledge Base	ReSeqTB	Curated TB mutation database with phenotypic correlation data
COMBAT-TB / TB-MODS	GitHub: COMBAT-TB	South African TB bioinformatics tools and reference datasets
Nanopore TB Sequencing Validation	Votintseva et al., J Clin Microbiol (2017)	Early validation of nanopore sequencing for TB resistance prediction
ONT TB Application Note	nanoporetech.com/resource-centre	Search for "tuberculosis" — ONT application notes with protocols
EQA for TB WGS	GenomeTrakr / WHO GLASS	Emerging EQA programs for genomic TB surveillance

5. FluoroCycler XT

Instrument Overview

Item	Detail
Manufacturer	Bruker / Hain Lifescience
Models	FluoroCycler XT
Use Cases	RT-PCR (FluoroType assays), Line Probe Assays (GenoType MTBDRplus, MTBDRsl)
Export Format	CSV from FluoroSoft

Software Downloads

Software	Version	Link	Notes
FluoroSoft	2.x	Bruker/Hain Lifescience	Contact Bruker/Hain representative. Bundled with instrument.

Registration note: FluoroSoft is not publicly downloadable; it is provided with the FluoroCycler XT instrument and updated through Bruker service.

Export Instructions

Detailed export instructions to be added after FluoroCycler XT field mapping specification is complete.

General path: FluoroSoft → Results → Export → CSV

Two distinct export types depending on assay:

1. **FluoroType assays (RT-PCR):** Ct-based results similar to QuantStudio export
2. **GenoType assays (LPA):** Band pattern interpretation results — unique format with probe presence/absence and resistance interpretation

Published Validation Data & References

Resource	Link	Description
Hain Lifescience GenoType MTBDRplus Package Insert	Provided with kit	Band interpretation tables, sensitivity/specificity data

Resource	Link	Description
Hain Lifescience GenoType MTBDRsl Package Insert	Provided with kit	Second-line drug resistance interpretation
WHO Policy: MTBDRplus/sl	WHO Molecular TB Diagnostics	WHO recommendation and performance data for line probe assays
Nathavitharana et al. (2017) — LPA Systematic Review	PubMed: PMC5470001	Meta-analysis of GenoType MTBDRplus/sl accuracy

6. SeqStudio Genetic Analyzer

Instrument Overview

Item	Detail
Manufacturer	Thermo Fisher Scientific / Applied Biosystems
Models	SeqStudio, SeqStudio Flex
Use Cases	HIV drug resistance genotyping, HCV genotyping, TB targeted sequencing
Export Format	.ab1 (raw) → bioinformatics pipeline → CSV/JSON to OpenELIS

Software Downloads

Software	Version	Link	Notes
SeqStudio Software	1.x / 2.x	ThermoFisher Connect Downloads	Instrument control and .ab1 file generation. ThermoFisher Connect account required.
Sequencing Analysis Software	6.x / 7.x	ThermoFisher Downloads	For viewing/editing .ab1 trace files.
Minor Variant Finder	2.x	Same portal	For detecting mixed bases / minor populations (relevant for drug resistance).

HIV Drug Resistance Pipeline Software:

Software	Version	Link	Notes
Stanford HIVdb Algorithm	Current	hivdb.stanford.edu	Gold standard for HIV drug resistance interpretation. Web-based and API available.
Sierra Web Service (HIVdb API)	2.x	GitHub: hivdb/sierra-graphql	Open source GraphQL API for programmatic HIVdb access.
RECall (basecalling/assembly)	Current	recall.bccfe.ca	BC Centre for Excellence HIV drug resistance basecalling tool.
HyDRA (PHAC pipeline)	Current	GitHub: phac-nml/hydra	Public Health Agency of Canada HIV DR pipeline.

Export Instructions

SeqStudio → .ab1 files:

1. Complete sequencing run in SeqStudio Software
2. Navigate to **Results → Export**
3. Export .ab1 trace files for each sample
4. Transfer .ab1 files to bioinformatics workstation

.ab1 → HIV Drug Resistance Report (via Stanford HIVdb):

1. Assemble forward/reverse .ab1 traces into consensus sequence (using RECall or manual alignment)
2. Submit consensus FASTA to Stanford HIVdb
3. Download structured JSON report (or use API)
4. Import JSON into OpenELIS

Detailed export and pipeline instructions to be added after SeqStudio field mapping specification is complete.

Published Validation Data & References

Resource	Link	Description
Stanford HIVdb Validation	hivdb.stanford.edu/page/algorithm-validation	Validation datasets and algorithm performance data
WHO HIV Drug Resistance Surveillance Guidelines	WHO HIVDR	Guidance on genotyping for DR surveillance

Resource	Link	Description
VQA (Virology Quality Assurance) Program	VQA	Proficiency testing for HIV genotyping laboratories
ANRS HIV Resistance Algorithm	hivfrenchresistance.org	Alternative resistance interpretation algorithm
Sanger Sequencing for HIV DR: WHO Protocol	WHO Manual	Standard protocol for HIV DR genotyping

7. KingFisher Flex

Instrument Overview

Item	Detail
Manufacturer	Thermo Fisher Scientific
Models	KingFisher Flex, KingFisher Apex
Use Cases	Nucleic acid extraction (pre-analytical); sample tracking & audit trail
Export Format	CSV log from BindIt Software

Software Downloads

Software	Version	Link	Notes
BindIt Software	4.x	ThermoFisher Downloads	Protocol design and run logging. ThermoFisher Connect account required.
KingFisher Instrument Control	Bundled	Included with BindIt	Firmware and hardware control.

Export Instructions

Menu Path: BindIt Software → Run Log → Export

1. Complete extraction run in BindIt Software
2. Navigate to **Run History** or **Log**

3. Select the completed run
4. Click **Export** → CSV
5. The log contains: Sample ID, Protocol Name, Plate Position, Start Time, End Time, Status, Reagent Lot, Operator

Key note: KingFisher exports are **pre-analytical tracking data**, not analytical results. The CSV links the extraction event (sample ID, protocol, reagent lot, operator, timestamps) to downstream testing for chain-of-custody audit trail.

Detailed export instructions to be added after KingFisher field mapping specification is complete.

Published Validation Data & References

Resource	Link	Description
KingFisher Application Notes	thermofisher.com/kingfisher	Extraction efficiency validation for various sample types
MagMAX Extraction Kit Validation	Provided with kit	Recovery rates, carry-over data, lot-specific QC

General Resources

Cross-Instrument References

Resource	Link	Description
CLSI AUTO11 — LIS Interface Standard	CLSI.org	Clinical Laboratory Standards Institute standard for instrument-LIS communication
ASTM E1394-97 — Standard Specification for Transferring Information Between Clinical Instruments and Computer Systems	ASTM.org	The ASTM protocol standard used by MALDI and other instruments
WHONET Software	whonet.org	Free microbiology lab database software for AMR surveillance; includes organism/antibiotic code files
WHONET Organism Code File	whonet.org/resources	Master organism code lookup table for WHONET export mapping

Resource	Link	Description
WHONET Antibiotic Code File	whonet.org/resources	Master antibiotic code lookup table
WHO GLASS	who.int/glass	Global Antimicrobial Resistance and Use Surveillance System
HL7 FHIR Laboratory Resource	hl7.org/fhir/diagnosticreport	FHIR standard for laboratory result reporting

Vendor Support Portals

Vendor	Portal	Covers
ThermoFisher Connect	thermofisher.com/connect	QuantStudio, Attune, SeqStudio, KingFisher
Bruker Care	bruker.com/services	MALDI Biotyper, FluoroCycler XT
Oxford Nanopore Community	community.nanoporetech.com	MinION, MinKNOW, Dorado

Version History

Version	Date	Changes
1.0	2025-02-19	Initial release covering all 7 instruments